

IDENTIFICATION

CODE : GI-3-S1-EC-PSX
ECTS : 4

HOURS

Cours : 2h
TD : 50h
TP : 4h
Projet : 0h
Evaluation : 4h
Face à face pédagogique : 60h
Travail personnel : 20h
Total : 84h

ASSESSMENT METHOD

IE (interrogation écrite)
IE (interrogation écrite)
IE (interrogation écrite)
EE+EO (évaluation écrite+orale)

TEACHING AIDS

Lecture handout and exercises
(part Probability & statistics and
part Design of experiments)
Use of spreadsheet files.
Course materials available
electronically.

TEACHING LANGUAGE

French
English

CONTACT

M. FONDREVELLE Julien :
julien.fondrevelle@insa-lyon.fr

AIMS

This course belongs to teaching unit Informatique et mathématiques décisionnelles (GI-3-S1-UE-IMAD) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 3)
- A2 Operate a model of a real or virtual system (Level 3)
- A3 Implement an experimental approach (Level 3)
- A5 Process data (Level 3)
- A6 Report on an analysis or on a scientific approach (Level 3)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
- C3 Evaluating, prototyping and simulating a system (Level 2)
- C8 Managing supply in connection with the planning and inventory management policy (Level 2)
- C10 Developing and implementing an action plan as part of a quality and continuous improvement approach (Level 2)
- C15 Managing a project based on a master plan (planning, budgeting, definition of monitoring indicators) (Level 1)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Notion of probability, Principles of combinatorial analysis, Probability calculus (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Notions of conditional probabilities and independent events (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Notion of random variable, Main indicators for random variables: expected value, variance... (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Main probability distributions (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Central limit theorem (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Notions of sample, population, statistical inference (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Client and supplier risks (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Ponctual estimation, Confidence intervals (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- One-tailed and two-tailed statistical tests, Comparison tests, goodness-of-fit tests, independence tests (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Implementation of a design of experiments (C10)

To be able to :

- Computing the probability of a complex event, based on simple events or by combinatorial analysis (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Modelling a random phenomenon choosing the appropriate probability distribution (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Computing and interpreting the main indicators related to a random variable (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Determining the average behaviour of a phenomenon considering a large number of random trials (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Carrying out a statistical test to validate a hypothesis related to a population (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Estimating the parameters of a statistical population based on a sample of values (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Carrying out a statistical test to validate a hypothesis related to a population (A1, A2, A3, A5, A6, C1, C3, C8, C10)
- Planning the tests to be conducted (A2, A3, C15)
- Carrying out the measurements (A1, C1)
- Developing a computing tool to process a design of experiments (A3)

Knowledge :

- Notion of probability, Principles of combinatorial analysis, Probability calculus
- Notions of conditional probabilities and independent events
- Notion of random variable, Main indicators for random variables: expected value, variance...
- Main probability distributions
- Central limit theorem
- Notions of sample, population, statistical inference
- Client and supplier risks
- Ponctual estimation, Confidence intervals
- One-tailed and two-tailed statistical tests, Comparison tests, goodness-of-fit tests, independence tests
- Optimal organization of experimentation
- Regression model and mathematical predictor
- Statistical model associated with the analysis of variance
- Implementation of a design of experiments

Capacities :

- Computing the probability of a complex event, based on simple events or by combinatorial analysis

- Modelling a random phenomenon choosing the appropriate probability distribution
- Computing and interpreting the main indicators related to a random variable
- Determining the average behaviour of a phenomenon considering a large number of random trials
- Computing the client / supplier risks according to a quality control procedure
- Estimating the parameters of a statistical population based on a sample of values
- Carrying out a statistical test to validate a hypothesis related to a population
- Planning the tests to be conducted
- Carrying out the measurements
- Developing a computing tool to process a design of experiments
- Analyzing the results obtained
- Finding an optimal setup condition

CONTENT

The aim of this course is to introduce the engineering tools and techniques needed to model and analyse phenomena and systems involving uncertainties, in particular through experimental approaches such as design of experiments. It is based on a description of standard mathematical concepts, coming from, among others, probability theory, descriptive statistics and analysis of variance. Numerous examples and application exercises are provided for each part of the course.

The knowledge and skills acquired are also put into practice through a case study based on a quality control problem, enabling the application of estimation methods and hypothesis testing, and through the implementation of an experiment for setting up a machine (a catapult), with a Taguchi's fractional design.

These different activities also lead the students to use and develop data processing tools on spreadsheets.

BIBLIOGRAPHY

D. Ghorbanzadeh : Probabilités - Exercices corrigés. Editions Technip. 1998.

Y. Dodge : Premiers pas en statistique. Springer. 2003.

S. Morgenthaler : Introduction à la statistique. Presses Polytechniques et Universitaires Romandes. 1997.

P. Bogaert : Probabilités pour scientifiques et ingénieurs: Introduction au calcul des probabilités, 2ème édition. De Boeck Supérieur. 2020.

G. Sado, M.C. Sado : Les plans d'expériences. De l'expérimentation à l'assurance qualité, Nlle édition. AFNOR. 2000.

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J. Goupy : Introduction aux Plans d'expériences, 2ème édition. Dunod. 2001.

G. Lasnier : Plans d'expériences en gestion industrielle. Hermes science publ. - Lavoisier. 2003.

PRE-REQUISITES

None

INSA LYON

Campus LyonTech La Doua

20, avenue Albert Einstein - 69621 Villeurbanne cedex - France

Tél. + 33 (0)4 72 43 83 83 - Fax + 33 (0)4 72 43 85 00

www.insa-lyon.fr

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